

Mathematical Funditions of Reinforcment Learning and Online Decision Making

Homework Assignment 2

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1 Question 1

1.1 a)

Yes. We use exactly the linear system, $v = R + Pv$, with $v(4) = 0$ and $R(4) = 0$.

1.1.1 Policy $\mu_1(s) = 1$: always choose action 1

The transition probabilities give

$$\begin{aligned}v_1 &= -1 + 0.9v_1 + 0.1v_2, \\v_2 &= -1 + 0.9v_2 + 0.1v_3, \\v_3 &= -1 + 0.9v_3 + 0.1v_4.\end{aligned}$$

Since $v_4 = 0$,

$$v_3 = -1 + 0.9v_3 \Rightarrow v_3 = -10.$$

Then

$$v_2 = -1 + 0.9v_2 + 0.1(-10) \Rightarrow v_2 = -20.$$

Finally,

$$v_1 = -1 + 0.9v_1 + 0.1(-20) \Rightarrow v_1 = -30.$$

Therefore,

$$V_{\mu_1} = (-30, -20, -10, 0).$$

1.1.2 Policy $\mu_2(s) = 2$: always choose action 2

Now the equations are

$$\begin{aligned}v_1 &= -2 + 0.5v_1 + 0.5v_2, \\v_2 &= -2 + 0.5v_1 + 0.5v_3, \\v_3 &= -2 + 0.5v_2 + 0.5v_4.\end{aligned}$$

Since $v_4 = 0$,

$$v_3 = -2 + 0.5v_2.$$

Solving the three equations gives

$$v_1 = -24, \quad v_2 = -20, \quad v_3 = -12.$$

Therefore,

$$V_{\mu_2} = (-24, -20, -12, 0).$$

1.2 b)

Comparing the two available actions at each state gives the optimal policy $\mu^* = (2, 2, 1)$.

The corresponding equations are

$$\begin{aligned}v_1 &= -2 + 0.5v_1 + 0.5v_2, \\v_2 &= -2 + 0.5v_1 + 0.5v_3, \\v_3 &= -1 + 0.9v_3.\end{aligned}$$

With $v_4 = 0$, solving gives

$$V_{\mu^*} = (-22, -18, -10, 0).$$

1.3 c)

Define the Bellman operator by $T(v) = R + Pv$. For any $v, w \in \mathbb{R}^{S+1}$, $T(v) - T(w) = P(v - w)$. Then

$$\|T^k(v) - T^k(w)\|_\infty \leq \|P\|_\infty^k \|v - w\|_\infty = \|v - w\|_\infty.$$

Thus T^k is not a contraction in the l_∞ -norm. might be in the weighted norm in Question 3, Section c.

2 Question 2

2.1 a)

Since state 0 is absorbing, $P(0, 0) = 1$ and $P(0, j) = 0$ for all $j \neq 0$. Therefore, the first row of P is $(1, 0, \dots, 0)$.

The remaining rows describe transitions from states $1, \dots, S$. We can separate transitions to state 0 into P_1 and transitions between states $1, \dots, S$ into P_2 .

Hence,

$$P = \begin{pmatrix} 1 & 0 \\ P_1 & P_2 \end{pmatrix},$$

where $P_1 \in \mathbb{R}^{S \times 1}$, $P_2 \in \mathbb{R}^{S \times S}$, and $0 \in \mathbb{R}^{1 \times S}$.

2.2 b)

Suppose, for contradiction, that $I - P_2$ is singular. Then there exists $x \neq 0$ such that $(I - P_2)x = 0$. Thus, $P_2x = x$. Hence, for any $k \geq 1$, $P_2^k x = x$.

Since there are finitely many states and state 0 is reachable from every state, there exists k such that

$$P^k(i, 0) > 0, \quad i = 1, \dots, S.$$

Thus P_2^k is strictly substochastic, so $\|P_2^k\|_\infty < 1$. But

$$\|x\|_\infty = \|P_2^k x\|_\infty \leq \|P_2^k\|_\infty \|x\|_\infty < \|x\|_\infty,$$

which is a contradiction. Therefore, $I - P_2$ is nonsingular.

2.3 c)

Using parts (a) and (b), we have

$$P = \begin{pmatrix} 1 & 0 \\ P_1 & P_2 \end{pmatrix}, \quad v = \begin{pmatrix} v(0) \\ \hat{v} \end{pmatrix}, \quad r = \begin{pmatrix} 0 \\ \hat{r} \end{pmatrix}.$$

The Bellman equation is $v = r + Pv$. Since state 0 is absorbing and $r(0) = 0$, $v(0) = 0$. For the remaining states, $\hat{v} = \hat{r} + P_2\hat{v}$. Therefore, $(I - P_2)\hat{v} = \hat{r}$. From part (b), $I - P_2$ is nonsingular, so $\hat{v} = (I - P_2)^{-1}\hat{r}$.

Hence,

$$v = \begin{pmatrix} v(0) \\ \hat{v} \end{pmatrix} = \begin{pmatrix} 0 \\ (I - P_2)^{-1}\hat{r} \end{pmatrix}.$$

3 Question 3

3.1 a)

Using the structure from Question 2(a),

$$P = \begin{pmatrix} 1 & 0 \\ P_1 & P_2 \end{pmatrix}, \quad r = \begin{pmatrix} 0 \\ \hat{r} \end{pmatrix}.$$

Also, since $v_k(0) = 0$ for every k , we can write $v_k = \begin{pmatrix} 0 \\ \hat{v}_k \end{pmatrix}$. The value iteration algorithm is $v_{k+1} = r + Pv_k$. Substituting the above expressions gives

$$\begin{pmatrix} 0 \\ \hat{v}_{k+1} \end{pmatrix} = \begin{pmatrix} 0 \\ \hat{r} \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ P_1 & P_2 \end{pmatrix} \begin{pmatrix} 0 \\ \hat{v}_k \end{pmatrix}.$$

Multiplying the matrices,

$$\begin{pmatrix} 0 \\ \hat{v}_{k+1} \end{pmatrix} = \begin{pmatrix} 0 \\ \hat{r} + P_2\hat{v}_k \end{pmatrix}.$$

Therefore, $\hat{v}_{k+1} = \hat{r} + P_2\hat{v}_k$. Thus the original value iteration is equivalent to iterating only over the S non-absorbing states, with any initialization $\hat{v}_0 \in \mathbb{R}^S$.

3.2 b)

Let

$$R'(s') = \begin{cases} 1, & s' \in S^+, \\ 0, & s' = 0. \end{cases}$$

Starting from state s , the value of this new MRP is

$$N(s) = \mathbb{E}_s \left[\sum_{k=0}^{\tau_0-1} 1 \right],$$

which is the expected number of visits to non-absorbing states before reaching 0.

By the Bellman equation,

$$N(s) = 1 + \sum_{s' \in S} P(s, s')N(s').$$

Since $N(0) = 0$, this becomes

$$N(s) = 1 + \sum_{s' \in S^+} P(s, s')N(s').$$

3.3 c)

Let $w_1, w_2 \in \mathbb{R}^S$. Since

$$B(w_1)(s) - B(w_2)(s) = \sum_{s' \in S^+} P(s, s')(w_1(s') - w_2(s')).$$

By the definition of the weighted norm, $|w_1(s') - w_2(s')| \leq N(s')\|w_1 - w_2\|_N$. Therefore,

$$\begin{aligned} |B(w_1)(s) - B(w_2)(s)| &\leq \sum_{s' \in S^+} P(s, s')N(s')\|w_1 - w_2\|_N \\ &= (N(s) - 1)\|w_1 - w_2\|_N, \end{aligned}$$

where the last equality follows from part (b). Dividing by $N(s)$,

$$\frac{|B(w_1)(s) - B(w_2)(s)|}{N(s)} \leq \frac{N(s) - 1}{N(s)}\|w_1 - w_2\|_N.$$

Taking the maximum over $s \in S^+$,

$$\|B(w_1) - B(w_2)\|_N \leq \alpha\|w_1 - w_2\|_N,$$

where

$$\alpha = \max_{s \in S^+} \frac{N(s) - 1}{N(s)}.$$

Since $N(s) \geq 1$, we have $\alpha \geq 0$. Also, because the expected time to absorption is finite, $N(s) < \infty$, so

$$\frac{N(s) - 1}{N(s)} < 1.$$

Since there are finitely many states, $\alpha < 1$. Hence B is an α -contraction.

Since v is the fixed point of B , we have $B(v) = v$. The value iteration algorithm is $v_{k+1} = B(v_k)$. Therefore,

$$\begin{aligned} \|v_{k+1} - v\|_N &= \|B(v_k) - B(v)\|_N \\ &\leq \alpha\|v_k - v\|_N. \end{aligned}$$

Applying this inequality repeatedly gives

$$\|v_k - v\|_N \leq \alpha^k\|v_0 - v\|_N.$$

Since $0 \leq \alpha < 1$, we have $\alpha^k \rightarrow 0$. Hence

$$\lim_{k \rightarrow \infty} \|v_k - v\|_N = 0.$$

Thus, the value estimation algorithm converges to v at the linear rate α .

3.4 d)

From part (c), we have

$$\|v_k - v\|_N \leq \alpha^k \|v_0 - v\|_N,$$

where

$$\alpha = \max_{s \in S^+} \frac{N(s) - 1}{N(s)} = 1 - \frac{1}{N_{\max}}, \quad N_{\max} := \max_{s \in S^+} N(s).$$

We want

$$\|v_k - v\|_N \leq \epsilon.$$

It is therefore sufficient that

$$\alpha^k \|v_0 - v\|_N \leq \epsilon.$$

Taking logarithms gives

$$k \geq \frac{\log(\|v_0 - v\|_N / \epsilon)}{-\log \alpha}.$$

Since

$$-\log \alpha = -\log \left(1 - \frac{1}{N_{\max}}\right) \geq \frac{1}{N_{\max}},$$

we obtain the sufficient condition

$$k \geq N_{\max} \log \left(\frac{\|v_0 - v\|_N}{\epsilon} \right).$$

Therefore, after

$$k = \left\lceil N_{\max} \log \left(\frac{\|v_0 - v\|_N}{\epsilon} \right) \right\rceil$$

iterations, we have

$$\|v_k - v\|_N \leq \epsilon.$$

4 Question 4

4.1 a)

Since $v = (I - \gamma P)^{-1} R$ and $\hat{v} = (I - \gamma P)^{-1} \hat{R}$, we have

$$\hat{v} - v = (I - \gamma P)^{-1} (\hat{R} - R).$$

Therefore,

$$\|\hat{v} - v\|_{\infty} \leq \|(I - \gamma P)^{-1}\|_{\infty} \|\hat{R} - R\|_{\infty}.$$

Let $B = -\gamma P$. Since P is a transition matrix, $\|P\|_{\infty} = 1$, and hence

$$\|B\|_{\infty} = \gamma < 1.$$

Using the given result,

$$\|(I - \gamma P)^{-1}\|_{\infty} \leq \frac{1}{1 - \gamma}.$$

Since $\|\hat{R} - R\|_{\infty} \leq \epsilon$, we obtain

$$\|\hat{v} - v\|_{\infty} \leq \frac{1}{1 - \gamma} \epsilon.$$

4.2 b)

Since v is the true value function, it is a fixed point of the Bellman operator: $v = B(v)$.

Therefore,

$$\|\hat{v} - v\|_\infty = \|\hat{v} - B(v)\|_\infty \leq \|\hat{v} - B(\hat{v})\|_\infty + \|B(\hat{v}) - B(v)\|_\infty.$$

The Bellman operator is a γ -contraction, so

$$\|B(\hat{v}) - B(v)\|_\infty \leq \gamma \|\hat{v} - v\|_\infty.$$

Using the assumption

$$\|\hat{v} - B(\hat{v})\|_\infty \leq \epsilon,$$

we get

$$\|\hat{v} - v\|_\infty \leq \epsilon + \gamma \|\hat{v} - v\|_\infty.$$

Thus,

$$(1 - \gamma) \|\hat{v} - v\|_\infty \leq \epsilon,$$

and hence

$$\|\hat{v} - v\|_\infty \leq \frac{\epsilon}{1 - \gamma}.$$

4.3 c)

Since

$$v = (I - \gamma P)^{-1} R$$

and

$$\hat{v} = (I - \gamma \hat{P})^{-1} R,$$

we have

$$v = R + \gamma P v$$

and

$$\hat{v} = R + \gamma \hat{P} \hat{v}.$$

Therefore,

$$\begin{aligned} \|\hat{v} - v\|_\infty &= \|\gamma \hat{P} \hat{v} - \gamma P v\|_\infty \\ &\leq \gamma \|\hat{P}(\hat{v} - v)\|_\infty + \gamma \|(\hat{P} - P)v\|_\infty \\ &\leq \gamma \|\hat{P}\|_\infty \|\hat{v} - v\|_\infty + \gamma \|\hat{P} - P\|_\infty \|v\|_\infty. \end{aligned}$$

Since $\hat{P}$ is a transition matrix,

$$\|\hat{P}\|_\infty = 1.$$

Also, from $v = R + \gamma P v$,

$$\|v\|_\infty \leq \|R\|_\infty + \gamma \|v\|_\infty,$$

so

$$\|v\|_\infty \leq \frac{\|R\|_\infty}{1 - \gamma}.$$

Thus,

$$(1 - \gamma) \|\hat{v} - v\|_\infty \leq \gamma \epsilon \frac{\|R\|_\infty}{1 - \gamma}.$$

Hence,

$$\|\hat{v} - v\|_\infty \leq \frac{\gamma \epsilon \|R\|_\infty}{(1 - \gamma)^2}.$$

Is it not the same as the question? But if we assume $\|v\|_\infty \leq 1$, then it is achievable.

5 Question 5

For both discounted and shortest-path MRPs, the choice between dynamic programming and solving the linear system mainly depends on the number of states and the desired accuracy.

- **Dynamic programming is more efficient** when the number of transitions per state is small (i.e., P is sparse) and we only need an approximate solution. Each iteration is then relatively cheap, and we can stop once the desired accuracy is reached.
- **Solving the linear system is more efficient** when we need a highly accurate solution, since it gives the exact value (up to numerical precision) without requiring many iterations. It can also be preferable when P is dense and the number of iterations required by dynamic programming is large.
- **The two methods are comparable** when the number of states is small or moderate, and the dynamic programming algorithm requires only a small number of iterations to reach the desired accuracy.

For discounted MRPs, dynamic programming converges geometrically at rate γ , so it becomes slower as γ approaches 1.

For shortest-path MRPs, the convergence rate depends on the expected time to reach the absorbing state. Thus, dynamic programming can be slower when the expected time to absorption is large.

6 Question 6

For any two Q -tables Q_1, Q_2 , we have

$$\begin{aligned} |T(Q_1)(s, a) - T(Q_2)(s, a)| &= \gamma \left| \mathbb{E}_{s' \sim P(\cdot|s, a)} \left[\max_{a'} Q_1(s', a') - \max_{a'} Q_2(s', a') \right] \right| \\ &\leq \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} \left| \max_{a'} Q_1(s', a') - \max_{a'} Q_2(s', a') \right|. \end{aligned}$$

Using

$$\left| \max_{a'} Q_1(s', a') - \max_{a'} Q_2(s', a') \right| \leq \max_{a'} |Q_1(s', a') - Q_2(s', a')|,$$

we obtain

$$|T(Q_1)(s, a) - T(Q_2)(s, a)| \leq \gamma \sum_{s'} P(s'|s, a) \max_{a'} |Q_1(s', a') - Q_2(s', a')| \quad (1)$$

$$\leq \gamma \max_{s', a'} |Q_1(s', a') - Q_2(s', a')|. \quad (2)$$

Taking the maximum over (s, a) ,

$$\|T(Q_1) - T(Q_2)\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty.$$

Thus, T is a contraction with parameter $\gamma < 1$. By the contraction mapping theorem, T has a unique fixed point Q^* . Since the fixed point satisfies

$$Q^* = T(Q^*),$$

it is the optimal Q -table.

For value iteration,

$$Q_{k+1} = T(Q_k).$$

Since $Q^* = T(Q^*)$,

$$\begin{aligned}\|Q_{k+1} - Q^*\|_\infty &= \|T(Q_k) - T(Q^*)\|_\infty \\ &\leq \gamma \|Q_k - Q^*\|_\infty.\end{aligned}$$

Applying this repeatedly gives

$$\|Q_k - Q^*\|_\infty \leq \gamma^k \|Q_0 - Q^*\|_\infty.$$

Therefore, value iteration converges linearly to Q^* at rate γ . In particular, to obtain $\|Q_k - Q^*\|_\infty \leq \epsilon$, it is sufficient that

$$k \geq \left\lceil \frac{\log(\|Q_0 - Q^*\|_\infty / \epsilon)}{-\log \gamma} \right\rceil.$$

7 Question 7

7.1 a)

$$r = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The process alternates deterministically between states 1 and 2. Hence, starting from either state, the rewards alternate between 1 and -1 , so the long-run average reward is $g(1) = g(2) = 0$. Thus,

$$g = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Since $g(s) = 0$, the cumulative rewards starting from state 1 are $1, 0, 1, 0, \dots$, whose Cesàro limit is $1/2$. Starting from state 2, they are $-1, 0, -1, 0, \dots$, whose Cesàro limit is $-1/2$. Therefore,

$$h(1) = \frac{1}{2}, \quad h(2) = -\frac{1}{2},$$

$$\text{and } h = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

7.2 b)

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad r = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The stationary distribution π satisfies

$$\pi = \pi P, \quad \pi_1 + \pi_2 = 1.$$

Thus,

$$(\pi_1, \pi_2) = (\pi_2, \pi_1),$$

which gives

$$\pi_1 = \pi_2 = \frac{1}{2}.$$

Therefore,

$$\pi = \left(\frac{1}{2} \quad \frac{1}{2} \right).$$

The gain is the expected reward under the stationary distribution:

$$g = \sum_{s \in S} \pi(s)r(s) = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0.$$

Hence,

$$\pi = \left(\frac{1}{2}, \frac{1}{2} \right), \quad g = 0.$$

7.3 c)

$$v_\lambda(1) = 1 - \lambda + \lambda^2 - \lambda^3 + \dots = \frac{1}{1 + \lambda},$$

$$v_\lambda(2) = -1 + \lambda - \lambda^2 + \lambda^3 - \dots = -\frac{1}{1 + \lambda}.$$

Thus,

$$v_\lambda = \begin{pmatrix} \frac{1}{1 + \lambda} \\ -\frac{1}{1 + \lambda} \end{pmatrix}.$$

The gain is

$$g = \lim_{\lambda \rightarrow 1^-} (1 - \lambda)v_\lambda(s).$$

Hence,

$$g = \lim_{\lambda \rightarrow 1^-} \frac{1 - \lambda}{1 + \lambda} = 0.$$

Therefore, $g = 0$.

7.4 d)

$$g\mathbf{1} + (I - P)h = r,$$

where

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad I - P = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad r = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Suppose (g, h) and (g', h') are two solutions. Then

$$g\mathbf{1} + (I - P)h = r$$

and

$$g'\mathbf{1} + (I - P)h' = r.$$

Subtracting the two equations gives

$$(g - g')\mathbf{1} + (I - P)(h - h') = 0.$$

Writing this out component-wise,

$$\begin{pmatrix} g - g' \\ g - g' \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} h_1 - h'_1 \\ h_2 - h'_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Multiplying the matrices gives

$$\begin{pmatrix} g - g' + (h_1 - h'_1) - (h_2 - h'_2) \\ g - g' - (h_1 - h'_1) + (h_2 - h'_2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Adding these two equations, the h -terms cancel:

$$2(g - g') = 0.$$

Hence,

$$g = g'.$$

Therefore, all solutions of the system have the same value of g .

To compute g , consider the original system:

$$\begin{pmatrix} g + h_1 - h_2 \\ g - h_1 + h_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Adding the two equations gives

$$2g = 1 + (-1) = 0.$$

Thus,

$$\boxed{g = 0}.$$

8 Question 8

8.1 a)

Since $g^*\mathbf{1} + h^* = r + Ph^*$, we have $h^* = r + Ph^* - g^*\mathbf{1}$. Therefore,

$$\begin{aligned} h_{k+1} - h^* &= r + Ph_k - (r + Ph^* - g^*\mathbf{1}) \\ &= P(h_k - h^*) + g^*\mathbf{1}. \end{aligned}$$

Adding a constant to every component does not change the span, so

$$\begin{aligned} \text{sp}(h_{k+1} - h^*) &= \text{sp}(P(h_k - h^*) + g^*\mathbf{1}) \\ &= \text{sp}(P(h_k - h^*)) \\ &\leq \gamma \text{sp}(h_k - h^*). \end{aligned}$$

Applying this recursively gives

$$\text{sp}(h_k - h^*) \leq \gamma^k \text{sp}(h_0 - h^*).$$

8.2 b)

Assume

$$\text{sp}(h - h^*) \leq \epsilon.$$

Define

$$c = \frac{\max_{s \in S} (h(s) - h^*(s)) + \min_{s \in S} (h(s) - h^*(s))}{2}.$$

Now define

$$e = h - h^* - c\mathbf{1}.$$

Therefore,

$$h = h^* + c\mathbf{1} + e.$$

Let

$$M = \max_{s \in S} (h(s) - h^*(s)), \quad m = \min_{s \in S} (h(s) - h^*(s)).$$

By the definition of c ,

$$c = \frac{M + m}{2}.$$

Hence, the maximum value of e is

$$\max_s e(s) = M - c = M - \frac{M + m}{2} = \frac{M - m}{2},$$

while the minimum value of e is

$$\min_s e(s) = m - c = m - \frac{M + m}{2} = -\frac{M - m}{2}.$$

Thus, the maximum and minimum of e are symmetric around zero. Therefore,

$$\begin{aligned} \|e\|_\infty &= \max_s |e(s)| \\ &= \frac{M - m}{2}. \end{aligned}$$

Since

$$\text{sp}(h - h^*) = M - m,$$

we obtain

$$\|e\|_\infty = \frac{1}{2} \text{sp}(h - h^*).$$

Using the assumption

$$\text{sp}(h - h^*) \leq \epsilon,$$

we get

$$\|e\|_\infty \leq \frac{\epsilon}{2}.$$

Therefore, there exists a scalar c and a vector e such that

$$h = h^* + c\mathbf{1} + e, \quad \|e\|_\infty \leq \frac{\epsilon}{2}.$$

8.3 c)

Since

$$g^k = r + Ph^k - h^k$$

and

$$g^*\mathbf{1} = r + Ph^* - h^*,$$

we obtain

$$\begin{aligned} g^k - g^*\mathbf{1} &= (r + Ph^k - h^k) - (r + Ph^* - h^*) \\ &= (P - I)(h^k - h^*). \end{aligned}$$

From part (b), choose

$$c = \frac{\max_s (h^k(s) - h^*(s)) + \min_s (h^k(s) - h^*(s))}{2}$$

and define

$$e = h^k - h^\star - c\mathbf{1}.$$

Then

$$h^k - h^\star = c\mathbf{1} + e$$

and, by the construction in part (b),

$$\|e\|_\infty = \frac{1}{2} \text{sp}(h^k - h^\star).$$

Since $P\mathbf{1} = \mathbf{1}$,

$$\begin{aligned} g^k - g^\star\mathbf{1} &= (P - I)(c\mathbf{1} + e) \\ &= (P - I)e. \end{aligned}$$

Therefore,

$$\begin{aligned} \|g^k - g^\star\mathbf{1}\|_\infty &= \|(P - I)e\|_\infty \\ &\leq \|Pe\|_\infty + \|e\|_\infty \\ &\leq 2\|e\|_\infty \\ &= \text{sp}(h^k - h^\star). \end{aligned}$$

Using part (a),

$$\text{sp}(h^k - h^\star) \leq \gamma^k \text{sp}(h_0 - h^\star).$$

Hence,

$$\|g^k - g^\star\mathbf{1}\|_\infty \leq \text{sp}(h_0 - h^\star)\gamma^k.$$

It is tighter than the bound, the question wants.